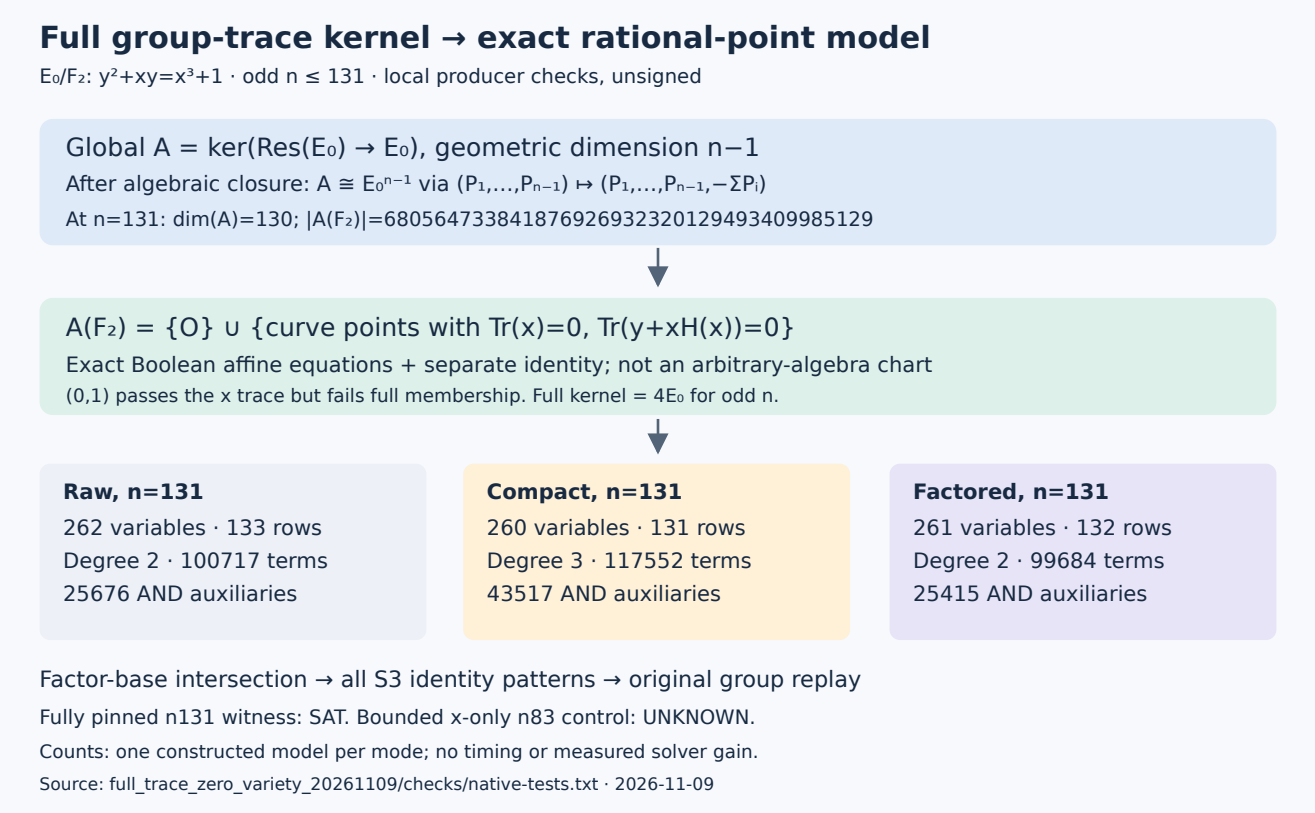
Full trace-zero variety: production integration

2026-11-09 · Isogeny-labs/crypto · PR3744 · local unsigned validation

Result: Full group-trace membership, projection, coordinates, three actual Boolean polynomial models, factor-base intersection and native S3 relation integration are implemented. Ten focused tests and the root library type-check pass locally. These are producer assertions with unverified independent standing.



Establishing the algebraic object

Let $k=F_2$, $K=F_{2^n}$, E_0/k: $Y^2Z+XYZ=X^3+Z^3$ and $W=\mathrm{Res}(K/k)(E_0)$. Define the scheme-theoretic kernel A of the descended group-sum map $W \to E_0$. After algebraic closure, W is E_0^n and the map sums the factors. The map from E_0^{n-1} sending the last coordinate to minus the sum of the others is an isomorphism onto A. Thus A is a smooth connected abelian variety of dimension $n-1$. A splitting over k is not asserted.

Its k-points are exactly the kernel T_n of the Frobenius point trace. $E_0(F_2)$ is cyclic of order four, and trace restricts to multiplication by odd n, so it is onto. The unique nontrivial geometric two-torsion point $(0,1)$ has nonzero trace, so the kernel has odd order. Therefore $T_n=4E_0(K)$, of size $\#E_0(K)/4$. This is standard restriction-of-scalars mathematics plus the written argument; no new formal proof or independent signed verdict is supplied.

Exact rational-point equations

The existing production halving derivation gives, for affine curve points, membership exactly when Tr(x)=0 and Tr(y+xH(x))=0. Here H is the odd-degree half-trace. A half exists iff lambda²+lambda=x has a solution; choose H(x). Its abscissa u satisfies u²=y+(lambda+1)x, so the second bit is exactly Tr(u). This includes the missing second halving test. The identity is separate.

In the polynomial basis, write Tr(x)=sum($t_i x_i$) with t_0=1, and q(x)=Tr(xH(x)). q is quadratic Boolean ANF. Expand the original curve into n coordinate rows. The raw model adds both trace rows. The compact model substitutes x_0 from the first bit and y_0 from the second, raising the degree bound to three. The factored model retains a helper h=q(x), keeping degree two.

n / mode	Variables	Rows	Terms	Degree	AND aux
83 raw	166	85	53707	2	10292
83 compact	164	83	63644	3	20468
83 factored	165	84	53042	2	10127
131 raw	262	133	100717	2	25676
131 compact	260	131	117552	3	43517
131 factored	261	132	99684	2	25415

These are constructed counts from one model per row, not solver timings. The compact model removes two bits but adds terms and auxiliaries. All variables are Boolean; equations live in $F_2[v]/(v_i^2+v_i)$. Together with the identity branch, they represent all F_2-rational points. They are not a positive-dimensional geometric defining ideal or an arbitrary-algebra chart. Those remain distinct software deliverables.

ECC2K-130 uses n=131, polynomial $z^{131}+z^{13}+z^2+z+1$ and full trace-kernel order 680564733841876926932320129493409985129. This is already the challenge subgroup. The dimension130 and two-bit coordinate elimination do not shrink that group.

Production path and guarantees

TraceZeroVariety validates curve shape, irreducibility and full cardinality; it supplies exact point trace, idempotent projection P−[n mod4]Tr(P), the different cofactor image tau(P)−P, and a checked codec. The factor-base API preserves exactly the supplied points in T_n and reduces the subspace basis by its x-trace bit. It does not silently replace them with cofactor images. The root index-calculus adapter uses this API.

The native BigUint S3 path now admits n=131, attaches full membership and curve equations to every summand and replays full points. The oracle handles all intermediate-identity patterns for arity2 through9 and lifts x-tuples to actual base points before original-group replay. Only complete exhaustion returns Refuted; resource caps return Inconclusive. Existing CLI selection is unchanged: these are explicit library entry points.

Checks and retained incomplete searches

The focused facade imports actual production field, group, model, S3 and SAT modules with no arithmetic mocks. Ten tests pass. Checks cover exhaustive point traces, projection and codecs at n3/5/7/9 (kernel sizes1/11/29/127), all model assignments at n3/5, actual n83/131 polynomial models on public fixtures and torsion cosets, malformed admission and allocation caps, factor-base intersection, complete n5 relation comparisons, certificate replay and changed-statement rejection, and the following wide controls.

The degree131 full-membership SAT encoding rejects x=0, whereas the x-trace bit alone admits the order-two point. Fully pinned P+(-P)+P=P controls give SAT with original-group replay including an identity intermediate. At n83: 31540 variables, 92296 clauses, 421 XORs. At n131:78076,230560,661. These check consistency of those witness encodings, not natural relation finding.

The separately bounded x-only-pinned n83 control returns **UNKNOWN** at its one-second/2000-conflict cap. Its test passes by retaining that explicit outcome; UNKNOWN is not a solution. An earlier unbounded control was interrupted with SIGINT after90+ seconds; its output is preserved. No natural degree131 relation, measured solving degree, yield, performance gain or challenge recovery is reported.

Root command cargo check –lib –no-default-features passes. This does not clear the earlier whole-root test failures or unrelated historical benchrec gate. Matched experiment execution still needs task-specific checkpoint/S3 credentials; no EKS or paid job ran. This work is local development validation.

Reproduction and evidence

README.md gives the exact locked cargo test/run commands and variable ordering. The native exporter emits actual ECC2K-130 ANF when given a new output path; otherwise it prints summary metadata. Infinity is explicit outside the affine equations. checks/native-tests.txt and checks/root-cargo-check.txt retain output; checks/commands.txt pins commands/environment; SOURCES.sha256 pins source. The generated large equation export is not committed.

The integral telescoping identity has certificate IDC1h3fcd7aa52221c17e, 32 issued plus32 replay points over $2^{61}-1$. It is not a characteristic-two membership certificate or formal theorem. Local evidence remains unsigned; no independent PCC standing is asserted.

Flow: full models and local checks completed; unbounded wide control interrupted; bounded control completed UNKNOWN; natural matched relation finding, performance and full DLP not attempted. Existing performance graphs retain their values; this study adds construction.svg only.

Sources

Gorla–Massierer arXiv:1405.1059, section2 (trace-zero variety), and1403.0126 (representation caveats); Kosters–Yeo1503.08001; Galbraith–Granger–Merz–Petit, SAC2020, Index Calculus; production curve.rs, is_in_4e and its point-halving references. The prior trace_zero_relation_check_20261109 study preserves the matched x-only layout audit. KB connector and executable are unavailable (wrapper exit127); these primary and repository sources are the fallback.